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Infection	Pathogens	1 st line therapy
Acute otitis media	<i>H. Influenzae, S. pneumoniae, M. Catarrhalis</i>	- Amoxicillin SD (45-60 mg/kg/day) in 3 divided doses if no risk of resistant <i>S. pneumoniae</i> - Amoxicillin HD 75-90 mg/kg/day in 2 divided doses if risk factors present - After 2-3 days of therapy and no improvement, switch to amoxicillin/clavulanate
Osteomyelitis of hematogenous source	Neonates: <i>group B streptococci, gram-negative enterics, S. aureus</i>	- MRSA unlikely: cloxacillin + cefotaxime - Possibly MRSA: vancomycin + cefotaxime or ceftriaxone
	Children: <i>S. aureus (MSSA and MRSA), group A streptococci, K. kingae (in <3 years)</i>	- MRSA unlikely: cloxacillin (avoid in <3 because <i>K. kingae</i> is resistant to clindamycin, vancomycin, and cloxacillin) or cefazolin – usually DOC (note, if a child is not immunized against <i>H. influenzae</i>, replace cefazolin with cefotaxime) - Possibly MRSA: vancomycin
	Adults: <i>S. aureus, gram-negative enterics</i>	- MRSA unlikely: cloxacillin or cefazolin - Possibly MRSA: vancomycin
Osteomyelitis due to spread from contiguous sites	Spread from head or neck: <i>S. aureus, gram-negative pathogens, mixed infection</i>	Penicillin + metronidazole or ceftriaxone + clindamycin
	Spread from soft tissue infection: <i>S. aureus, streptococci</i>	- MRSA unlikely: cloxacillin or cefazolin - Possibly MRSA: vancomycin
	Spread from genitourinary system: gram-negative enteric bacilli	Fluoroquinolone, 3 rd generation cephalosporin or extended-spectrum penicillin
Osteomyelitis due to penetrating trauma (e.g. foot puncture wound)	<i>P. aeruginosa, S. aureus</i>	- Children: - Cloxacillin or cefazolin + ceftazidime; may alternatively use an extended-spectrum penicillin + beta-lactamase inhibitor (e.g. piperacillin + tazobactam) - Adults: fluoroquinolone or a combination of an extended-spectrum penicillin + beta-lactamase inhibitor

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Infection	Pathogens	1 st line therapy
Mild-moderate <i>C. difficile</i> diarrhea	<i>C. difficile</i>	Fidaxomicin 200mg PO BID x 10 days OR (preferred) Vancomycin 125mg PO QID x 10 days OR Metronidazole 500mg PO TID x 10 days (NOT preferred)
Severe <i>C. difficile</i> diarrhea		Vancomycin 125mg PO QID x 10 days (preferred) OR Fidaxomicin 200mg PO BID x 10 days
Severe + complicated <i>C. difficile</i> diarrhea	<i>C. difficile</i>	Vancomycin 500mg PO QID +/- Metronidazole 500mg IV q8h (may consider adding rectal vancomycin if ileus)
Recurrent <i>C. difficile</i> diarrhea (recurrence within 8 weeks of completing treatment)	<i>C. difficile</i>	First recurrence: If fidaxomicin, vancomycin or metronidazole used in 1 st episode, use tapering/pulsed dose vancomycin
Bacterial meningitis	Age < 1 month: <i>S. agalactiae</i> , <i>E. coli</i> , <i>L. monocytogenes</i> , <i>Klebsiella species</i>	Ampicillin + cefotaxime OR ampicillin + aminoglycoside
	Age 1-23 months: <i>S. pneumoniae</i> , <i>N. meningitidis</i> , <i>S. agalactiae</i> , <i>H. influenzae</i> , <i>E. coli</i>	Vancomycin + 3 rd generation cephalosporin (ceftriaxone or cefotaxime). Add ampicillin if immunocompromised
	Age 2-50 years: <i>N. meningitidis</i> , <i>S. pneumoniae</i> , <i>H. influenzae</i>	Vancomycin + 3 rd generation cephalosporin (ceftriaxone or cefotaxime) Add ampicillin if immunocompromised
	Age > 50 years: <i>S. pneumoniae</i> , <i>N. meningitidis</i> , <i>L. monocytogenes</i> , aerobic gram-negative bacilli	Vancomycin + ampicillin + 3 rd generation cephalosporin (ceftriaxone or cefotaxime)
Pneumonia (outpatient)	<i>S. pneumoniae</i>	Amoxicillin or doxycycline or macrolide (azithromycin or clarithromycin)
Pneumonia (outpatient with comorbidities or inpatient without comorbidities)	<i>S. Pneumoniae</i>	Respiratory quinolone OR Beta-lactam (amoxicillin/clavulanate or cephalosporin are preferred) + macrolide

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Infection	Pathogens	1 st line therapy
Urinary tract infection: cystitis	<i>E. coli</i> , <i>S. saprophyticus</i> , <i>P. mirabilis</i> , <i>K. pneumoniae</i> , <i>Enterococcus faecalis</i>	Any 1 of the following: - SMX/TMP - Trimethoprim - Nitrofurantoin - Fosfomycin
Urinary tract infection: pyelonephritis	<i>E. coli</i> , <i>P. mirabilis</i> , <i>K. pneumoniae</i>	Mild-moderate: fluoroquinolone (i.e. ciprofloxacin or levofloxacin) Severe: aminoglycoside IV +/- ampicillin IV then step-down to PO
Urinary tract infection: complicated	<i>E. coli</i> , <i>P. Mirabilis</i> , <i>Enterococcus faecalis</i> , other, <i>K. pneumoniae</i>	Mild-moderate SMX/TMP OR amox/clav OR fluoroquinolone OR cefixime Severe: aminoglycoside IV +/- ampicillin IV then step-down to PO
Folliculitis/furuncle/carbuncle/abscess	<i>S. aureus</i>	- Folliculitis: topical mupirocin or fusidic acid (PO cephalexin if no improvement) - Furuncle or carbuncle with systemic symptoms e.g. fever: - MSSA: cephalexin or cloxacillin - MRSA: SMX/TMP or doxycycline
Non-purulent cellulitis	Group A streptococcus, <i>S. aureus</i>	- Mild-moderate: cephalexin - Severe: cefazolin IV OR clindamycin PO/IV (if MRSA, give SMX/TMP, doxycycline, or vancomycin IV)
Purulent cellulitis	Group A streptococcus, <i>S. aureus</i>	- Mild-moderate: cephalexin OR clindamycin (if MRSA, give SMX/TMP OR amoxicillin+doxycycline) - Severe: cefazolin OR cloxacillin OR clindamycin IV (if MRSA, give vancomycin, linezolid, or daptomycin)
Pharyngitis	Group A beta-hemolytic streptococcus	Penicillin or amoxicillin
Sinusitis	Mostly viral (rhinovirus, influenza, parainfluenza), sometimes bacterial in children (<i>S. pneumoniae</i> , <i>H. influenzae</i> , <i>M. catarrhalis</i>)	Amoxicillin +/- clavulanate

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